TRINEXIS: An Alert-Aware Narration Policy and a Protocol for Measuring Its Effect on Alert Delivery

Yug G. Lakhani[1]
[1]Department of Computer Engineering, Gyanmanjari Innovative University, Bhavnagar, Gujarat, India
Corresponding author: Yug G. Lakhani, yglakhani@gmiu.edu.in
ORCID: https://orcid.org/0009-0004-2930-4953

Abstract— Open-vocabulary webcam systems must decide which detected objects to speak when output is short and time-limited. This report frames an alert-aware narration policy and a protocol for measuring its effect on alert delivery; that effect remains unmeasured. The policy ranks user-designated alert objects before truncating the spoken shortlist. Confidence-first and nearest-object-first baselines provide two explicit comparison orderings. The research question is whether alert-first narration makes users notice a designated object faster or more often than confidence-first narration. TRINEXIS implements the policy in an interval-throttled, language-configured webcam system and adds optional real-log instrumentation, a frozen alert manifest, a 5-10 session replay harness, Flask route tests, and continuous integration. Experiments A-D specify hardware timing, labeled detection evaluation, a vocabulary/confidence grid, real-log alert replay, and checked-language ratings; all empirical result cells remain NO DATA. The report does not claim a measured user benefit, detection quality, latency, distance accuracy, multilingual comprehension, or assistive effectiveness. Its contribution is the policy, its failable software checks, and a preregistrable protocol that reports whatever alert-first effect is observed, including a null or negative one, not a description of a Flask application.

Keywords / Index Terms— alert-aware narration; open-vocabulary detection; research software; speech interface; empirical protocol; interval throttling; technical report

I. INTRODUCTION

A. Motivation and Research Question

Object detection predicts bounding boxes and semantic labels for objects in an image. The YOLO family formulated detection as a single regression problem [1], and COCO-style benchmarks standardized evaluation over a fixed vocabulary of 80 released categories [2]. Vision-language pre-training broadened the interface: CLIP enabled zero-shot transfer from natural-language supervision [3], Grounding DINO fused language and vision for open-set detection [4], and YOLO-World introduced a prompt-then-detect paradigm [5]. Ultralytics exposes that family through a set_classes interface [6]. These advances make configurable object names possible, but an interactive narration system still has to decide what to say first when only a few objects can be spoken.

The research question is: Does alert-first narration make users notice a designated object faster or more often than confidence-first narration? This report separates two measurable parts. Replayed real logs can measure whether an alert reaches the spoken shortlist and how long alert-to-speech delivery takes. Whether a person notices the object faster or more often requires a separate consented user study. Neither effect has been measured here; all empirical cells remain NO DATA.

TRINEXIS is the research instrument used to implement the policy: an interval-throttled YOLO-World webcam pipeline with a browser dashboard, language-configured speech output, session telemetry, and optional real-log instrumentation. Its maintained backend, app23.py, uses Flask control routes [7], OpenCV capture [8], Ultralytics YOLO [15], and gTTS speech [9]. The Flask application is implementation infrastructure, not the paper's contribution.

B. Contribution

The contribution is an alert-aware narration policy and a protocol for measuring its effect on alert delivery, not a Flask app or a general webcam-application description. User-designated alert objects are ranked before the spoken shortlist is truncated. The ordering is isolated in a dependency-free policy module, checked by mutation-sensitive unit tests, instrumented for real-log replay, and deliberately kept separate from empirical detection or user-benefit claims.

- A policy and research question: alert-first, confidence-first, and nearest-object-first narration for a predesignated object, with delivery and noticing treated as distinct endpoints.
- A maintained implementation with AGPL-3.0 licensing [24], configurable camera/security settings, optional real-log timing and speech-event recorders, and a frozen alert-manifest mode.
- Real Flask test-client route tests, deterministic policy tests, continuous integration, and a source-level evidence-boundary check.
- Preregistrable Experiments A-D with hardware/data/rating manifests, runnable harnesses, and blank NO DATA result templates.

C. Scope and Limitations

This manuscript is a standalone technical report with experiment protocols and blank empirical results; it is not submission-ready. No labeled detection dataset, hardware timing campaign, known-distance calibration, translated-template rating, consented user study, dashboard screenshot, or spoken input-to-output example was completed. The author selected arXiv/institutional-report formatting; the current arXiv Submission Guidelines, moderation, metadata, and content-type pages were read on 10 Nov 2026 and recorded in paper/venue_guide_notes.md. arXiv has no mandatory venue template, so the generic single-column layout and numeric citations are retained. The author confirmed no external funding, no competing interests, and the CRediT roles actually performed; the AI-tool disclosure was checked against arXiv's current text-to-text generative-AI policy. The author must still personally verify every number after experiments. No detection-quality, latency, distance-error, robustness, multilingual-comprehension, assistive-effectiveness, or user-benefit result is claimed.

II. RELATED WORK / LITERATURE REVIEW

A. Detection and Open-Vocabulary Models

The original YOLO paper evaluated PASCAL VOC 2007/2012 and artwork-domain generalization on the Picasso and People-Art datasets; it did not evaluate on COCO [1]. COCO later supplied a fixed-vocabulary benchmark [2]. CLIP [3], Grounding DINO [4], YOLO-World [5], OWL-ViT [11], GLIP [12], Detic [13], and YOLOE [14] represent successive approaches to language-conditioned or open-vocabulary recognition. Their published results were obtained under distinct datasets, splits, prompts, hardware, and protocols, so they are not treated here as comparative baselines for TRINEXIS. Table I summarizes scope and non-comparability.

TABLE I. VERIFIED RELATED WORK AND NON-COMPARABILITY

Source	Contribution	Verified evaluation context	Relevance and limitation
Redmon et al. [1], 2016	One-stage YOLO detector	PASCAL VOC 2007/2012; Picasso and People-Art artwork generalization	Established real-time closed-set detection; no COCO evaluation in that paper and a different hardware/protocol context
Lin et al. [2], 2014	Microsoft COCO benchmark	Approximately 328,000 images, 2.5 million labeled instances, 80 released categories	Standard fixed-vocabulary protocol; utils/coco.txt is only a label list in this repository
Radford et al. [3], 2021	CLIP contrastive pre-training	400 million image–text pairs; zero-shot transfer over more than 30 datasets	Representation learning, not a real-time bounding-box detector

Source	Contribution	Verified evaluation context	Relevance and limitation
Liu et al. [4], 2024	Grounding DINO, ECCV 2024 published version	Open-set and referring-expression benchmarks under published protocols	Alternative open-set approach; not used in this repository
Cheng et al. [5], 2024	YOLO-World and RepVL-PAN	CVPR 2024, pp. 16901–16911	Direct model-family basis; published benchmark results are not reproduced here
Minderer et al. [11]; Li et al. [12]; Zhou et al. [13]; Wang et al. [14]	OWL-ViT, GLIP, Detic, YOLOE	ECCV/CVPR/arXiv protocols with distinct vocabularies and datasets	Expanded open-vocabulary context; incompatible with this repository's prompts and runtime
Alqahtani et al. [19], 2024	Edge accuracy–latency–energy benchmark	YOLOv8, SSD, and EfficientDet Lite across heterogeneous edge devices	Motivates separate timing instrumentation; its measurements are not reused as TRINEXIS results

B. Assistive Interfaces, Speech, and Privacy

Remote sighted assistance has been studied through Be My Eyes [16], while Seeing AI [17] and AI scene-description applications [18] illustrate practical auditory presentation of visual information. These systems are relevant to the design space but do not validate TRINEXIS. The repository configures 72 language codes and 62 top-level speech resources. Ten language codes use an English template fallback in the UI; the remaining 72-language claim is unchecked. gTTS uses Google Translate's speech service [9], so raters should listen to sample audio and the author should listen to a few languages before any quality claim. Camera-based systems also raise privacy and face-recognition concerns [20], [21]. The optional face-recognition hook remains unavailable, and no consent, retention, or biometric-processing protocol is claimed.

C. Research Software and Reproducibility

Git LFS stores small pointer files in Git while large model content is held elsewhere [10]. The prior immutable software release [23] includes CITATION.cff, codemeta.json, dependency-license notes, pinned manifests, and a weights download script that verifies Git LFS object identifiers as SHA-256 checksums. The unreleased working-tree revision adds experiment protocols, route tests, CI, and instrumentation that are not part of that release. The project adopts AGPL-3.0-only [24] because Ultralytics YOLO is distributed under AGPL-3.0 [22]. A Zenodo DOI has not been minted; the final GitHub/Zenodo release must come last because a DOI is permanent for the exact version.

III. SYSTEM DESIGN

A. Architecture and Control Flow

Figure 1 shows the runtime and control flow. A browser client calls Flask control routes and displays a Motion JPEG stream. A configurable camera backend supplies frames to OpenCV. Detector calls are gated by an interval throttle, after which YOLO-World produces text-conditioned detection records. Those records feed visual overlays, a relative distance index, a time-decayed heatmap, session telemetry, an alert-aware speech selector, and optional real-log recorders. Optional face recognition is shown as a dashed, unavailable path. There is no training or benchmark-evaluation path.

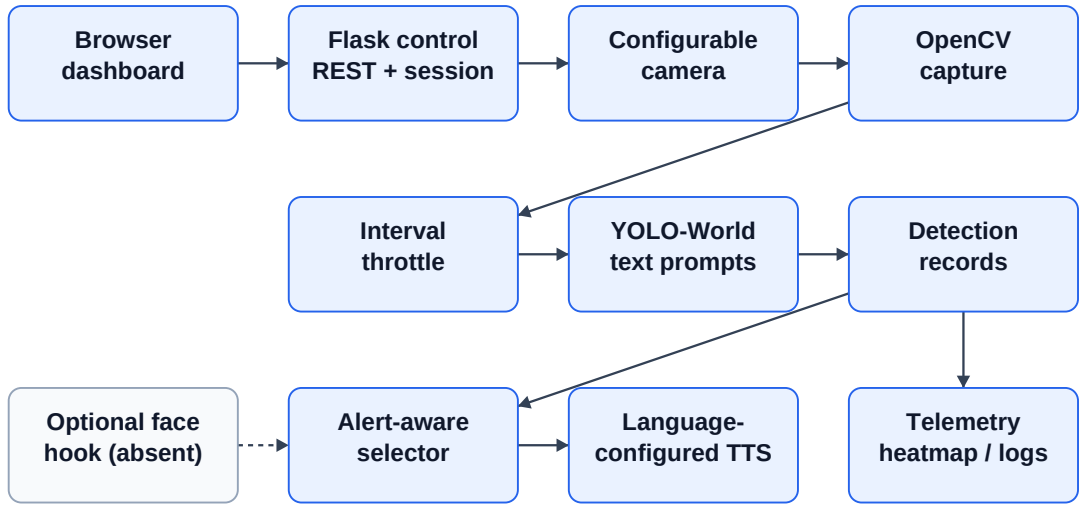

Fig. 1. TRINEXIS system architecture. Solid arrows denote implemented runtime paths; the dashed arrow denotes an optional/unavailable face-recognition hook. Vector SVG/PDF versions are provided in the release artifacts.

B. Interval Throttle and Alert-Aware Narration

Let t be the current capture time, t_prev the previous detector invocation, and Delta the configured throttle interval. A detector call is admitted when

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t - t_prev >= Delta. (1)
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The default is Delta = 1.5 s and the spoken shortlist is limited to two objects. Let A be the set of user-designated alert names, c(o) the detector confidence, and d(o) the relative distance index. The selector sorts every detection record o by the ascending key

```
R(o) = (I_alert(o), -c(o), d(o)), (2)
```

and only then takes the first two records. In Eq. (2), I_alert(o) = 0 when o.name is in A and 1 otherwise; alert objects therefore receive priority regardless of whether they are among the two highest-confidence detections. A confidence-first shortlist followed by alert-aware reranking inside the shortlist would not provide this invariant. For Experiment C, alert names A are declared in a manifest before collection; the runtime freezes /set_alerts and embeds the manifest hash in detection and speech-event records.

C. Time-Based Heatmap and Relative Distance Index

Eq. (3) replaces the older per-frame 0.995 multiplier, whose decay depends on camera frame rate, with elapsed wall-clock time dt and an adjustable half-life tau:

$$H_t = 2^{(-dt/\tau)} * H_{(t-1)} + \sum_i C(c_i; 35). \quad (3)$$

In Eq. (3), H_t is the heatmap at time t, C(c_i; 35) is a filled circle of radius 35 pixels centered on detection box c_i, and tau defaults to 60 s. The 60 s value is an adjustable default chosen to retain visible activity for roughly one minute; it is not fitted to data. The distance overlay is a unitless, relative size-based index

$$d_{idx} = (W * F) / \max(w_{px}, 1), \quad (4)$$

where w_px is the observed box width, W = 60, and F = 360. W and F are heuristic constants, not calibrated physical measurements; d_idx has no metric unit and must not be interpreted as centimetres or metres.

D. Configuration, Security, Language Fallback, and Instrumentation

Camera index, backend, resolution, host, port, optional API token, read-only mode, single-client mode, client timeout, session secret, and heatmap half-life are read from environment variables. The default bind address is 127.0.0.1 rather than 0.0.0.0. /set_model accepts only a checked-in model allowlist. Mutating control endpoints can require an X-API-Token or Bearer header: missing or invalid tokens return 401, unsupported schemes and read-only mutations return 403, and a second controlling client receives 409. Capture-frame and detector-invocation rates are reported separately. Optional JSONL recorders remain off unless TRINEXIS_LATENCY_LOG and TRINEXIS_EVENT_LOG are configured; they timestamp capture, inference, TTS synthesis, playback, rendering, speech enqueue, and playback events. /get_languages exposes the English-fallback language list and an empty verified-language list.

TABLE II. IMPLEMENTATION FEATURES AND VERIFICATION

Feature	Implementation	Verification
Open-source distribution	AGPL-3.0 LICENSE [24], README, pinned requirements, CITATION.cff, codemeta.json, and dependency-license notes	Source-contract tests and README instructions
Confidence display	Dashboard reports AVG CONFIDENCE rather than ACCURACY	Source-contract test
Alert-aware speech	Rank alerts before shortlist truncation via trinexis_core.select_speech_objects	Mutation-sensitive unit test and tests/ synthetic unit-test artifact
Client-state safety	Single-client control arbitration, 401/403/409 behavior, and stale-session timeout	Flask route tests and pure-function unit tests
Network defaults	Default host 127.0.0.1; optional API token and read-only mode on mutating control routes	Flask route tests and source-contract tests
Model loading	Allowlisted model filenames with HTTP 400 on rejection	Unit and route tests
Heatmap decay	Wall-clock exponential decay with adjustable half-life	Unit test
Distance display	Unitless relative distance index with explicit non-calibration note	Unit and source-contract tests
Camera portability	Configurable camera index and backend	Unit and source-contract tests
Rate reporting	Separate capture_fps and inference_fps counters and dashboard displays	Source-contract test
Language fallback	Ten missing templates fall back to English; 72 language codes remain unchecked	Route, language, and source-contract tests
Real-log instrumentation	Optional JSONL timing and speech-event recorders with manifest hashes	Telemetry unit tests and Experiment C protocol
Figures	Enlarged PNG plus standalone SVG and vector-PDF exports	make_updated_figures.py

IV. SOFTWARE VERIFICATION

A. Verification Environment

All checks were static, synthetic, or exercised through Flask's test client with mocked native modules. The preparation environment used Python 3.11.2 with the standard library and Flask 3.1.3 for route tests; manuscript and figure generation used the pinned packages in paper/requirements-paper.txt. No model weights were hydrated, no camera was opened, no live Flask server was run, no network speech request was made, and no human participant was involved. Figure 2 summarizes the workflow. GitHub Actions is configured to run the same test suite on every push; a local CI-equivalent run passed, but no empirical result follows from a green test suite.

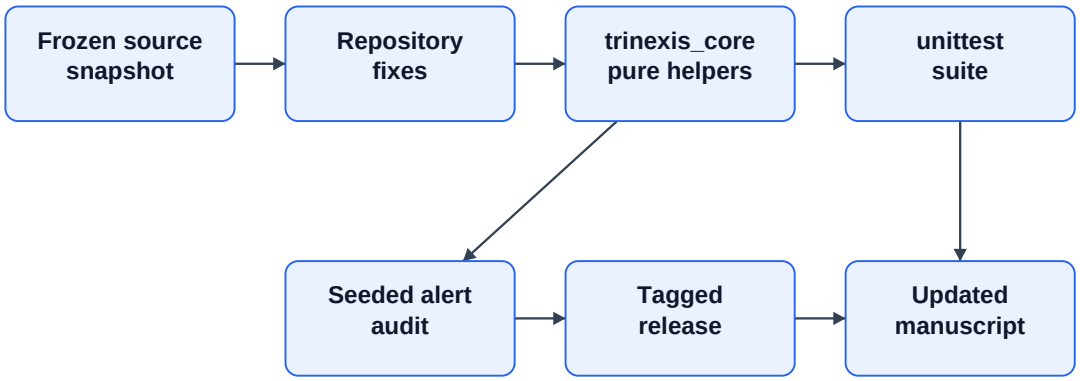

Fig. 2. Software-verification workflow. Every step is static, synthetic, or a mocked Flask route test; no model inference, camera capture, labeled dataset, or human data is used. Vector SVG/PDF versions are provided in the release artifacts.

B. Synthetic Alert-Selection Unit Test

The seeded 1,000-case synthetic check is generated only by tests/test_synthetic_regression.py; the old experiments/ generator was removed. Each case has six records, one alert object, and a two-object speech shortlist. With six records and a top-2 shortlist, a uniformly ranked alert falls outside the shortlist with probability 4/6 (66.7%). The confidence-first ordering omitted the alert object in 681 of 1000 cases (68.1%), consistent with that probability setup; the alert-first ordering omitted it in 0 of 1000 cases (0.0%). A deliberately broken confidence-first selector makes the unit test fail. This is a failable unit-test artifact of an ordering invariant, not an experiment: it does not show how often real detections contain alerts, whether a real user hears an alert, or whether speech output is useful.

TABLE III. SEEDED SYNTHETIC UNIT-TEST REGRESSION CHECK

Metric	Confidence-first	Alert-first	Change
Cases with a configured alert object	1,000	1,000	—
Alert omissions from the two-object shortlist	681	0	–681
Omission rate	68.1%	0.0%	–68.1 pp
Evidence type	Synthetic unit test	Synthetic unit test	No model inference

C. Route, Policy, and Source-Contract Tests

The standard-library unittest suite contains 93 tests: 43 + 37 + 13 = 93, namely 43 Flask route tests, 37 policy/harness/telemetry tests, and 13 source-contract tests. The exact CI statement is: 43 route tests plus 37 policy tests, run in CI. Route tests are reported separately from source-text tests. They exercise all 23 routes' status codes, JSON-object validation, clamping, API-token 401 behavior, unsupported-scheme/read-only 403 behavior, Bearer authentication, single-client 409 behavior, frozen alert manifests, and language-support metadata. Camera, NumPy, detector, and TTS imports are mocked with unittest.mock; no live runtime is validated by these counts. Policy tests cover narration ranking, mutation sensitivity, control status, telemetry arithmetic, language fallback, experiment-harness self-checks, and the seeded synthetic unit test. Continuous integration runs the suite on every push and pull request.

TABLE IV. TEST-SUITE SUMMARY

Test group	Tests	Failures	Errors
Flask test-client route tests	43	0	0
Policy/harness/telemetry tests	37	0	0
Repository/source-contract tests	13	0	0
Total	93	0	0

D. Release and Reproducibility

The prior immutable software release is GitHub release v0.3.0-standalone [23]. The working-tree fix revision described here is unreleased; it adds CITATION.cff/codemeta metadata updates, experiment protocols, real-log instrumentation, route tests, CI, language-fallback documentation, and the reframed paper. weights/download_weights.py parses the checked-in Git LFS pointers and verifies the declared SHA-256 object identifiers after Git LFS materialization. Appendix A lists reproduction commands, and Appendix B lists the principal artifacts.

E. Future Work

- Run Experiments A-D on declared hardware, labeled data, real logs, and checked languages; report negative results as-is.
- Obtain consent and institutional ethics approval before any user-noticing study.
- Calibrate the distance proxy at known distances or retain the unitless index with its current disclaimer.
- Add a dashboard screenshot and one spoken input-to-output example from a real hand-run smoke test; venue-policy and author-declaration checks are recorded, but author verification of final numbers remains.

V. EMPIRICAL PROTOCOLS AND BLANK RESULTS

A. Preregistration and Real Logs

All empirical cells are NO DATA. Before collection, the author must freeze hardware/software manifests, the paper and code, a 200-300-image report split, frozen prompt sets, and a predeclared alert-object manifest. The live app writes timestamped JSONL only when TRINEXIS_LATENCY_LOG and TRINEXIS_EVENT_LOG are configured. /export_logs supplies detection frames; event logs supply speech enqueue/playback records. The harnesses reject explicitly synthetic rows for empirical runs. Alert objects must be chosen in advance; the frozen manifest prevents post-hoc selection. No tuning on reported images is allowed, and the author must personally verify every number.

B. Experiments A-D

Experiment A collects at least 500 camera frames per setting at 416, 640, and 832 input sizes on declared CPU/GPU/RAM/OS, Python, camera, audio, and TTS-backend configuration. It reports median and p95 for capture, inference, TTS synthesis, playback, and rendering. Experiment B evaluates 200-300 labeled COCO, LVIS, or own images with pycocotools or Ultralytics val, reporting precision, recall, mAP@0.5, and speed; its grid is {455, 20, 50, 80} prompts x {0.15, 0.25, 0.35}. Experiment C declares alert objects in advance, runs 5-10 real webcam sessions with TRINEXIS_EVENT_LOG, and replays the same detection frames through alert-first, confidence-first, and nearest-object-first orderings. It reports delivery rate and mean alert-to-speech delay exactly as observed even if alert-first helps less than hoped. Experiment D samples 5-10 languages with native or reliable bilingual raters on 10 announcements each, rated 1-5 for correctness and naturalness; only checked languages may be reported, the remaining 72-language claims are unchecked, and raters should listen to gTTS/Google Translate speech samples. Negative or worse results are reportable.

The C harness is experiments/multi_session_replay.py. It rejects fewer than 5 or more than 10 sessions, explicitly synthetic rows, unmatched alert-manifest hashes, and missing speech-event logs. Its blank result cells are NO DATA; no session has been supplied. The optional E harness, experiments/user_study.py, generates only a blank counterbalanced worksheet and refuses to summarize without an institutional ethics approval identifier, documented consent, 8-12 real participants, all three conditions per participant, and numeric outcomes.

C. Blank Result Table

TABLE V. EMPIRICAL PROTOCOLS AND BLANK RESULTS

Experiment	Primary endpoint	Prerequisite	Result
A. Latency	Median/p95 by stage and input size	Declared hardware; 500+ frames per 416/640/832 setting	NO DATA
B. Detection	Precision/recall/mAP@0.5/speed	200-300 labeled images; separate tuning split; pycocotools or Ultralytics val	NO DATA
B. Grid	Twelve vocabulary/confidence cells	Frozen {455, 20, 50, 80} prompts x {0.15, 0.25, 0.35}	NO DATA
C. Alert replay	Delivery rate; mean alert-to-speech delay	Real /export_logs and speech events; predeclared alert manifest	NO DATA
D. Language	Mean correctness/naturalness	5-10 checked languages x 10 announcements; 1-5 ratings; listen to samples	NO DATA
E. User noticing	Notice rate; time-to-notice	Consent and institutional ethics approval; three-policy 8-12 person within-subject study	NO DATA

D. What Replay Does Not Establish

Experiment C measures alert delivery from logs, not whether a person notices a designated object faster or more often. A nearest-object-first comparison clarifies the ordering question but cannot substitute for a consented, ethics-approved user study with preregistered notice rate and time-to-notice outcomes. No such approval, participant, or user benefit is claimed. The dashboard screenshot and spoken input-to-output example are also pending. A final arXiv/institutional research article should be prepared only after these empirical steps are complete.

VI. DISCUSSION

A. Interpretation

The alert-aware policy makes the spoken shortlist independent of whether a designated object happens to be among the highest-confidence detections. Interval throttling separates detector cadence from camera frame rate; time-based heatmap decay and the unitless distance index remove two frame-rate- and calibration-dependent ambiguities. The Flask route tests establish control behavior under mocked dependencies, not live deployment success. Published detector literature is not comparable to this webcam pipeline because its datasets, prompts, splits, and hardware protocols differ. No measured effect on alert delivery or user noticing is reported.

B. Ethical and Practical Implications

Camera-based interfaces can expose bystanders and sensitive environments. Privacy, consent, retention, and biometric-processing policies must be defined before any real-world or assistive-use study [20], [21]. The AGPL-3.0 license [24] and dependency-license notice clarify redistribution obligations but do not substitute for an ethics review. The system should not be described as assistive technology for blind or low-vision users until a consented study demonstrates benefit.

C. Language Coverage and TTS

Seventy-two language codes and 62 top-level speech resources are configured. Ten language codes lack a top-level template and use the English fallback, which is now disclosed by /get_languages and the UI. No language is verified; no multilingual comprehension or translation-quality claim is made. gTTS uses Google Translate's speech service [9]. A checked-language study must use native or reliable bilingual raters, listen to sample audio, and report only the 5-10 languages actually rated; the author should also listen to a few languages. Missing templates may be added later only with reliable review; English fallback is documented rather than silently assumed.

VII. CONCLUSION AND FUTURE WORK

This report defines an alert-aware narration policy, implements it in a configurable open-vocabulary webcam system, and supplies a protocol for measuring its effect on alert delivery. Software verification establishes the ordering invariant, HTTP control behavior, language fallback, and blank-result harnesses. The measured effect remains unknown: detection evaluation, hardware timing, real-log alert replay, checked-language ratings, and any user-noticing study are future work. The final release and Zenodo DOI must follow completed evidence and author verification, not precede them.

DECLARATIONS

A. Funding and Competing Interests

The author confirmed no external funding and no competing interests on 10 Nov 2026. The author must re-verify these declarations before final submission.

B. Author Contributions (CRediT)

The author confirmed the roles actually performed: Yug G. Lakhani—conceptualization, software, and writing—original draft. Methodology, investigation, and formal analysis will be added only after those experiments are actually performed; validation, data curation, and review/editing must likewise match real work. No validation dataset or curated research data were used in this report.

C. AI-Tool Disclosure

Arena.ai Agent Mode, a text-to-text generative AI tool, was used for source auditing, harness/test scripting, figure generation, repository scripting, and manuscript formatting. This disclosure was checked against the current arXiv moderation policy, which requires significant text-to-text generative-AI use to be reported, places full responsibility on the named author, and does not permit an AI language tool to be listed as an author. The author must verify every number, reference, and declaration before submission. No institutional ethics approval is claimed.

D. Data and Code Availability

The prior immutable software release is available from GitHub release v0.3.0-standalone [23] under AGPL-3.0-only [24]; Ultralytics' AGPL-3.0 license is cited directly [22]. The unreleased working-tree revision on arena/69777d37-tech-yolo contains the reframed paper, three-policy experiment protocols, instrumentation, route tests, CI, and language-fallback UI; it is not part of that release. Model weights are Git LFS pointers and are fetched and checked with weights/download_weights.py [10]. No labeled dataset, experimental log, or valid Zenodo DOI is claimed. Zenodo DOI has not been minted because the final GitHub/Zenodo release must come last; the author authorized that sequence but has not yet connected Zenodo or approved the frozen v1.0.0 archive. After that final release, insert the DOI into CITATION.cff, README.md, Appendix A, and this Data and Code Availability section, then rebuild the manuscript. Dependency licenses are documented in THIRD_PARTY_LICENSES.md. The author selected arXiv/institutional-report formatting; official current arXiv policies were read and recorded in paper/venue_guide_notes.md, and no mandatory venue template exists, so the generic single-column PDF and numeric citations are retained. This manuscript is not submission-ready.

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APPENDIX A. REPRODUCTION COMMANDS

The branch below is mutable and represents the unreleased working-tree revision, not the prior immutable tag.

```
git clone https://github.com/Unusualyug/tech_yolo.git
cd tech_yolo
git checkout arena/69777d37-tech-yolo
python3 -m pip install -r requirements-ci.txt
python3 -m unittest discover -s tests -v
python3 -m unittest tests.test_synthetic_regression -v
python3 scripts/report_test_counts.py
python3 experiments/latency_benchmark.py --self-check
python3 experiments/detection_eval.py --self-check
python3 experiments/vocab_confidence_grid.py --self-check
python3 experiments/alert_replay.py --self-check
python3 experiments/language_eval.py --self-check
python3 experiments/multi_session_replay.py --self-check
python3 experiments/user_study.py --self-check
python3 scripts/live_smoke_test.py --self-check
python3 scripts/check_arxiv_metadata.py --check
python3 scripts/final_release_checklist.py --self-check
python3 scripts/verify_no_fabricated_results.py
# Live studies only; CI does not install these or open hardware:
python3 scripts/live_smoke_test.py --check-prereqs
python3 scripts/live_smoke_test.py --verify-evidence
python3 -m pip install -r requirements.txt
python3 weights/download_weights.py --fetch --verify
python3 -m pip install -r paper/requirements-paper.txt
python3 paper/make_updated_figures.py
python3 paper/build_updated_manuscript.py
python3 paper/audit_updated_repository.py
```

APPENDIX B. PRINCIPAL ARTIFACTS

- paper/tech_yolo_updated_research_paper.pdf, .docx, and .md: standalone technical report with blank empirical results.
- paper/artifacts/alert_selection_audit.json: tests/ unit-test artifact for the seeded synthetic alert-ordering regression.
- paper/artifacts/test_summary.json: route/policy/source-contract counts and exact CI report phrase.
- paper/artifacts/fig1_system_architecture_updated.{png,svg,pdf} and fig2_implementation_verification.{png,svg,pdf}: figure exports.
- paper/artifacts/updated_repository_audit.json: machine-readable working-tree/test summary.
- experiments/README.md and experiments/RESULTS_TEMPLATE.md: experiment protocols and blank NO DATA cells.
- experiments/latency_benchmark.py, detection_eval.py, vocab_confidence_grid.py, alert_replay.py, language_eval.py, and common.py: runnable harnesses.
- tests/test_routes.py, tests/route_test_support.py, and tests/test_synthetic_regression.py: real Flask route tests and the synthetic unit test.
- .github/workflows/ci.yml, requirements-ci.txt, scripts/report_test_counts.py, and scripts/verify_no_fabricated_results.py: CI and evidence-boundary checks.
- weights/download_weights.py and weights/weights_checksums.sha256: Git LFS pointer parser and SHA-256 manifest.
- zenodo.json and ZENODO.md: prepared Zenodo metadata and final-release DOI-minting instructions.

- paper/reference_verification.md: per-reference DOI/publisher verification log with all 24 rows ticked after recheck.
- paper/venue_guide_notes.md, experiments/LIVE_SMOKE_TEST.md, experiments/SPOKEN_IO_EXAMPLE.template.md, experiments/USER_STUDY_PROTOCOL.md, experiments/ethics_inquiry_receipt.md, experiments/session_manifest.template.json, experiments/user_study_worksheet.template.csv, experiments/multi_session_replay.py, experiments/user_study.py, scripts/live_smoke_test.py, scripts/check_arxiv_metadata.py, and scripts/final_release_checklist.py: venue, live-evidence, three-policy, ethics, and final-release preparation artifacts.
- FIX_STATUS.md, trinexis_core.py, trinexis_telemetry.py, trinexis_language.py, CITATION.cff, codemeta.json, README.md, LICENSE, requirements.txt, and THIRD_PARTY_LICENSES.md: software and status artifacts.